Multivariate Analysis 2025: Team Project

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```r
# We start by loading the libraries and the data for the implementations.

load("subjects-fMRI.RData")
source("brain_plots.R")
library(pheatmap)
library(gridExtra)
library(grid)
library(patchwork)
library(knitr)
```

1 Exploratory Data Analysis (EDA)

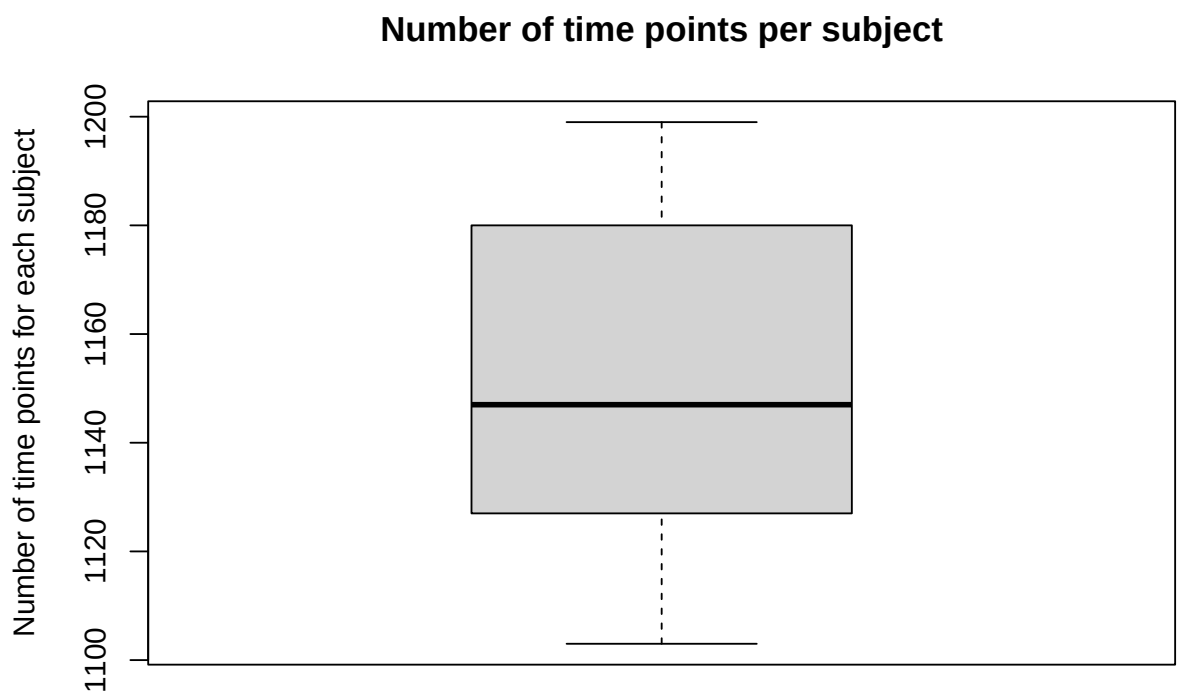
The `subjects-fMRI.RData` dataset is a subset of the WU-Minn HCP 1200 Subjects dataset (Glasser et al., 2013), preprocessed by MIPLab for multivariate analysis. Functional MRI (fMRI) measures brain activity through blood oxygenation level-dependent (BOLD) signals, which serve as proxies for neural activation in each region of interest (ROI).

The dataset includes 90 subjects (49 female, 41 male) aged 22–36+, distributed as follows: 15 aged 22–25, 37 aged 26–30, 37 aged 31–35, and 1 aged 36+. For each subject j, we have a data matrix $\mathbf{X}^{(j)}$ of dimension $N \times T_j$, where $N = 100$ ROIs and $T_j \approx 1150$ time points (varying slightly by subject). Each element $x_{n,t}^{(j)}$ represents the BOLD signal for ROI n at time t for subject j:

$$\mathbf{X}^{(j)} = \begin{bmatrix} x_{1,1}^{(j)} & \cdots & x_{1,T}^{(j)} \\ x_{2,1}^{(j)} & \cdots & x_{2,T}^{(j)} \\ \vdots & \ddots & \vdots \\ x_{N,1}^{(j)} & \cdots & x_{N,T}^{(j)} \end{bmatrix}.$$

```r
#summary(subj.x) # Summary of covariates (gender, age)

# Visualization (example for subject 5 at time 1 (t1),we performed this visualization in order to
↪  know how the output of the function brain plot
#brain_plot(subj.fmri[[5]][,1], roi.names,type = "diverging")

# We retrieve the number of time points for each subject and put them in a variable col_data in
↪  order to do a boxplot and highlight the variability.
col_data=c()
for (i in 1:length(subj.fmri)){
  col_data[i] =  length(subj.fmri[[i]][1,])
}

#col_data
#print(summary(col_data))
```

```r
boxplot(col_data,
        main = "Number of time points per subject",
        ylab ="Number of time points for each subject"
          )
```



The boxplot shows that T_j varies slightly across subjects (min = 1103, max = 1199, mean $\approx$ 1150).

Each ROI name (`roi.names`) corresponds to a specific functional network. The notation LH/RH indicates the left or right hemisphere. There are seven networks represented in total:

1. Default Mode (self-referential thought, memory)
2. Salience (switching between cognitive states, emotion)
3. Central Executive (working memory, decision-making)
4. Somatomotor (movement and sensory processing)
5. Visual (perceptual and visual processing)
6. Limbic (emotion and memory)
7. Dorsal Attention (goal-directed attention)

According to `roi.names`, ROIs are ordered by hemisphere and network:

- Left Hemisphere (1–50): Visual, Somatomotor, Dorsal Attention, Salience, Limbic, Central Executive, Default
- Right Hemisphere (51–100): Visual, Somatomotor, Dorsal Attention, Salience, Limbic, Central Executive, Default

For subsequent analyses, ROIs can be reordered by network to facilitate visualization and interpretation.

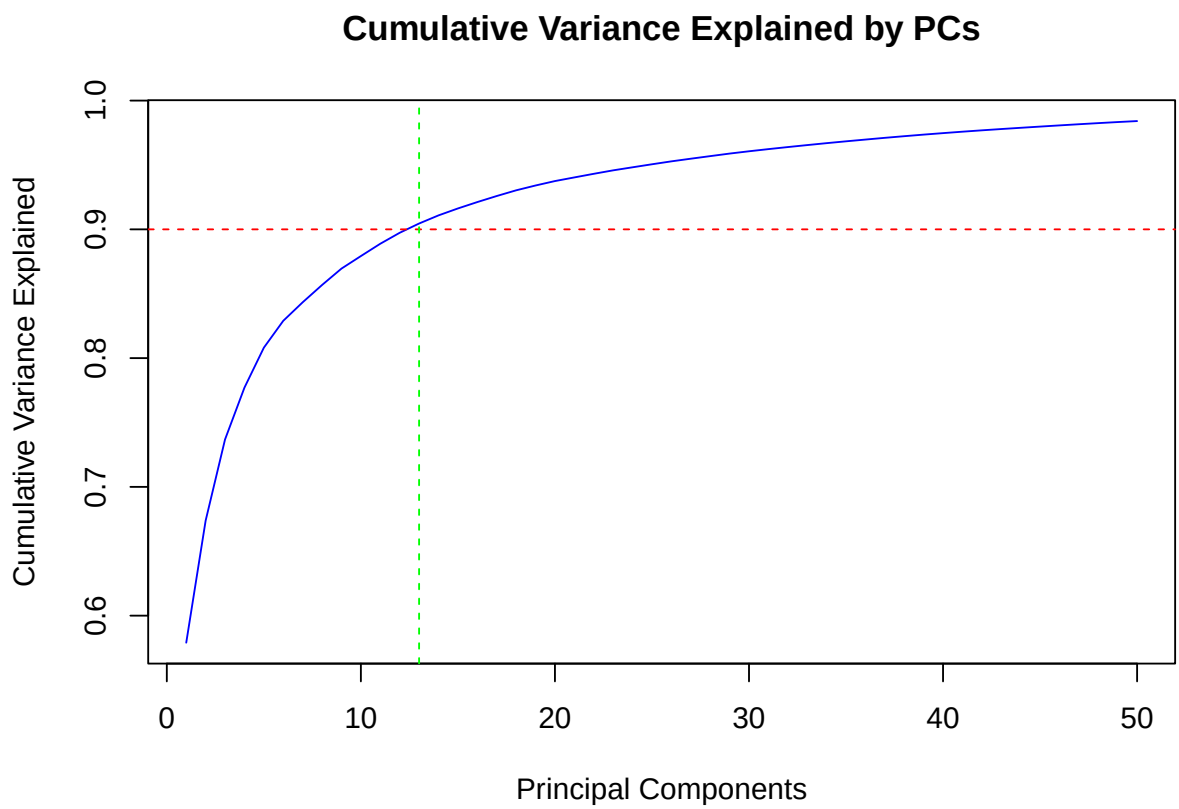
2 Dimensionality Reduction Using PCA (Single Subject Analysis)

We applied Principal Component Analysis (PCA) to subject $j = 5$ (female, aged 31–35), to reduce the high-dimensional fMRI data (100 ROIs $\times \sim 1150$ time points). PCA decomposes the data into orthogonal components summarizing variance across regions, where:

- Eigenvalues reflect the variance explained by each component.
- Loadings (rotation coefficients) indicate each ROI's contribution.

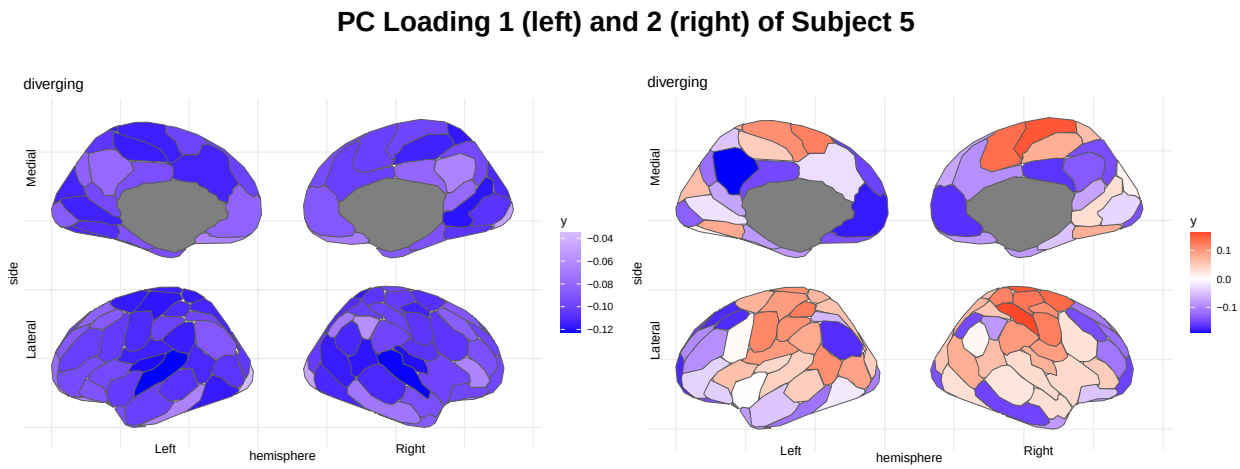
To determine the optimal number of PCs, we examined the cumulative variance explained.

```
# Check subject 5 gender and age
#subj.x[5,]

# Extract data for subject 5
subject_5_data <- subj.fmri[[5]]

# Verification of the dimension
#dim(t(subject_5_data))

#Perform PCA for subject 5
pca_result <- prcomp(t(subject_5_data), center = TRUE, scale. = TRUE)
#pca_result$rotation
#summary(pca_result)

# Scree plot to visualize the proportion of explained variance
explained_variance <- pca_result$sdev^2 / sum(pca_result$sdev^2)
#plot(explained_variance[1:30], type = "b", pch = 19, xlab = "PC", ylab = "Proportion of explained
↪  variance for the first 30 PCs", main = "Scree Plot - Subject 5")

cumulative_variance <- cumsum(explained_variance)

# Number of PC explaining more than 90% of cumulative variance
k_PC_90 = which(cumulative_variance >= 0.9)[1]

# Plot the cumulative variance for the first 50 PCs for convenience of the plot
plot(cumulative_variance[1:50], type = "l", col = "blue", xlab = "Principal Components",
     ylab = "Cumulative Variance Explained", main = "Cumulative Variance Explained by PCs")
abline(h = 0.9, col = "red", lty = 2) # Mark 90% variance line
abline(v = which(cumulative_variance >= 0.9)[1], col = "green", lty = 2)
```



The first 5 PCs explain ~ 80 of the total variance, while 13 PCs reach 90%. PC1 alone explains 57.9%, capturing a global activation pattern (all ROIs co-activate in the same direction). PC2 (9.48%) shows opposing activations between regions (positive vs. negative loadings), while PC3 (6.3%) reflects localized patterns.

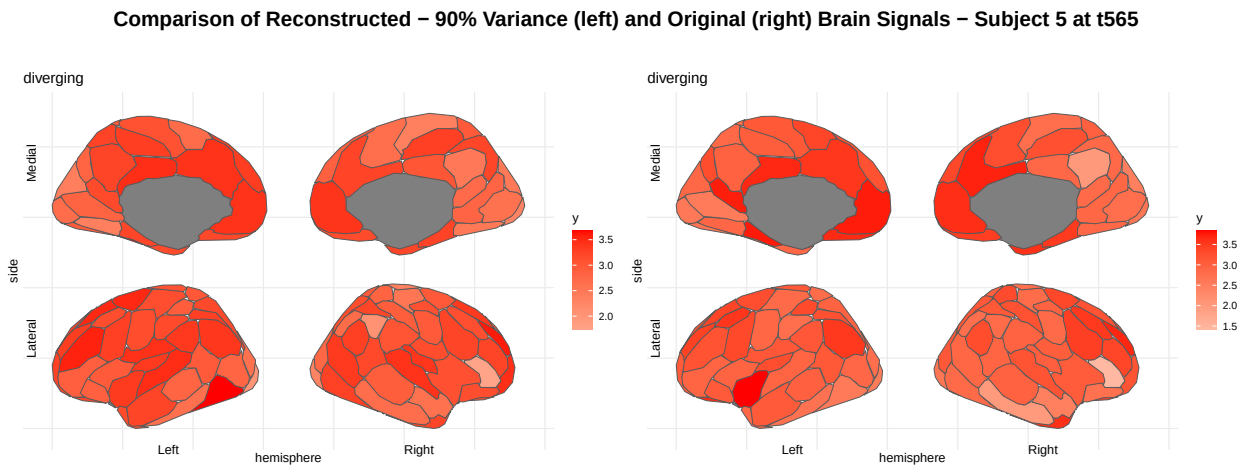
```
#Brain plot for PC loading 1 and PC loading 2 of subject 5
plot_pc1 = brain_plot(pca_result$rotation[, 1], roi.names,"diverging")

plot_pc2 = brain_plot(pca_result$rotation[, 2], roi.names,"diverging")

#plot_pc3 = brain_plot(pca_result$rotation[, 3], roi.names, type = "diverging")

grid.arrange(plot_pc1, plot_pc2, ncol = 2, widths = c(0.5, 0.5),
             top = textGrob("PC Loading 1 (left) and 2 (right) of Subject 5",
                            gp = gpar(fontsize = 24, fontface = "bold")))
```



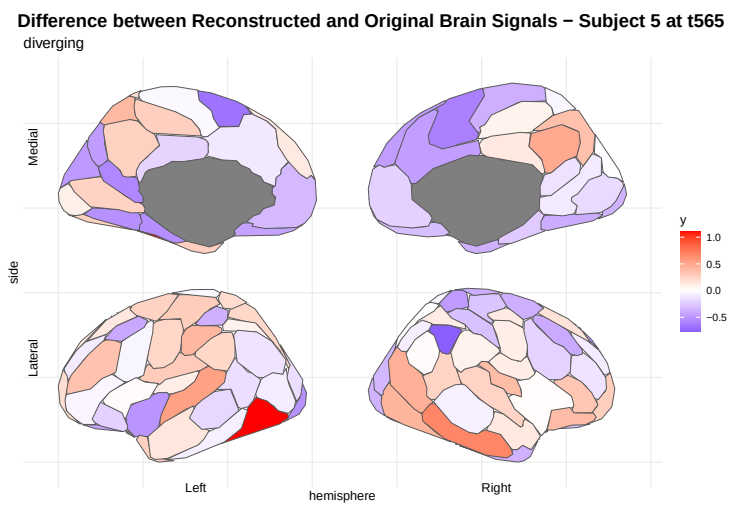
We then visualized brain activation at the peak time point $T_5 = 565$, corresponding to the maximum mean signal across ROIs. This moment represents the highest neural engagement for subject 5.

The PCA reconstruction (retaining 90% variance) closely replicates the original activation map, indicating PCA effectively preserves major activation patterns with limited components.

```
# Example of brain visualization the average maximum active time point
# Compute the mean of each row
row_means <- colMeans(subj.fmri[[5]])

# Find the row with the absolute maximum mean value
max_row_index <- which.max(abs(row_means))

# Get the maximum mean value for that row
max_row_value <- row_means[max_row_index]

# Print the result - at t = 565, j = 5 has the absolute maximum mean value regarding ROIs
↪  activation and de-activation
#cat("Row with the absolute maximum mean value:", max_row_index, "\n")

# Matrix reconstruction using 90% of explained variance
approx_5_90 <- pca_result$center + (pca_result$x[, 1:k_PC_90] %*%
↪  t(pca_result$rotation[,1:k_PC_90]))

# Set common color scale across both plots
global_min <- min(c(approx_5_90[max_row_index, ], subj.fmri[[5]][, max_row_index]))
global_max <- max(c(approx_5_90[max_row_index, ], subj.fmri[[5]][, max_row_index]))

# Generate brain plots for reconstructed and original signals
plot_pc_90 <- brain_plot(approx_5_90[max_row_index, ], roi.names,"diverging")
plot_original <- brain_plot(subj.fmri[[5]][, max_row_index], roi.names,"diverging")

# Arrange plots side by side
grid.arrange(plot_pc_90, plot_original, ncol = 2, widths = c(0.5, 0.5),
  top = textGrob("Comparison of Reconstructed - 90% Variance (left) and Original (right) Brain
  ↪  Signals - Subject 5 at t565",
                 gp = gpar(fontsize = 18, fontface = "bold"))
)
```




```r
# Compute the difference between original and reconstructed signals at t = 565
diff_approx_vs_orig <- approx_5_90[max_row_index, ] - subj.fmri[[5]][, max_row_index]

# Get min and max for symmetric color scale centered around 0
diff_limit <- max(abs(diff_approx_vs_orig))  # for symmetric diverging scale

# Generate the brain plot for the difference (error)
plot_diff <- brain_plot(diff_approx_vs_orig, roi.names,
                        "diverging")

# Display the difference plot
grid.arrange(plot_diff,
  top = textGrob("Difference between Reconstructed and Original Brain Signals - Subject 5 at t565",
                 gp = gpar(fontsize = 16, fontface = "bold")))
```



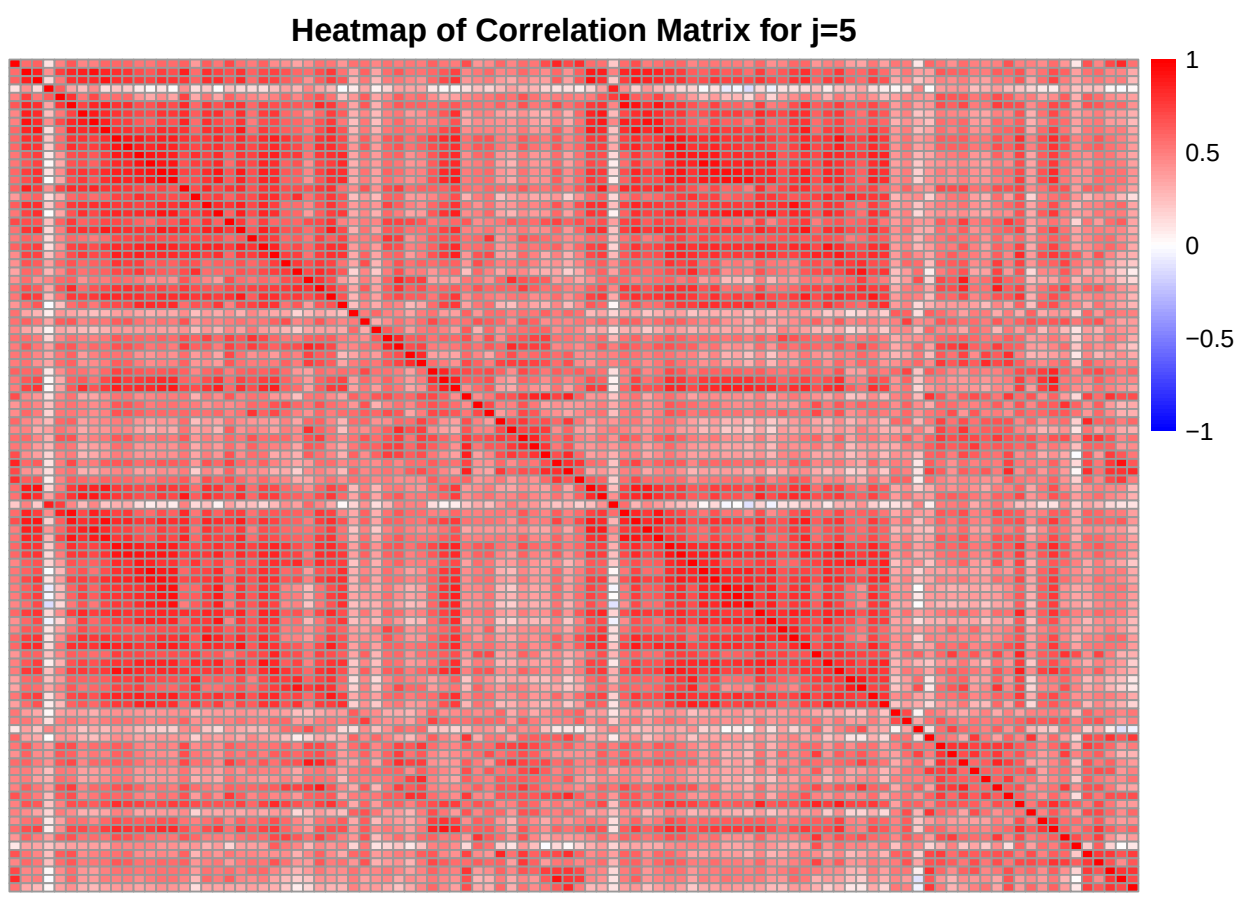

```r
# mse <- mean((approx_5_90[max_row_index, ] - subj.fmri[[5]][, max_row_index])^2)
# cat("Mean Squared Error (MSE) at t = 565:", round(mse, 4), "\n")
#
# mae <- mean(abs(approx_5_90[max_row_index, ] - subj.fmri[[5]][, max_row_index]))
# cat("Mean Absolute Error (MAE) at t = 565:", round(mae, 4), "\n")
```

3 Extending PCA to Multiple Subjects

Applying PCA directly to all subjects is not straightforward because: (1) the number of time points T_j differs across subjects, (2) time points are not temporally aligned, (3) functional variability exists between brains, and (4) data dimensionality ($N \gg$ number of subjects) makes direct PCA unstable.

To overcome these issues, we represent each subject by their **ROI–ROI correlation matrix** (100×100). This approach captures functional connectivity while avoiding temporal alignment problems. Each correlation matrix is then reshaped into a vector of length 10,000, forming a $90 \times 10{,}000$ dataset (one subject per row). Following Liégeois et al. (2017) and Liu et al. (2018), PCA on correlation matrices allows robust cross-subject comparison and dimensionality reduction.

```r
# Create correlation matrices and visualize subject j=5
# Example of the correlation matrix of subject j=5 via pheatmap visualisation.
color_palette <- colorRampPalette(c("blue", "white", "red"))(100)

cor_matrices <- lapply(subj.fmri, function(subj.fmri) cor(t(subj.fmri)))
vectorized_cor_matrix <- do.call(rbind, lapply(cor_matrices, as.vector))
corr_5 <- matrix(vectorized_cor_matrix[5,], nrow = 100, ncol = 100)
pheatmap(corr_5, color = color_palette,
         breaks = seq(-1,1,length.out=101),
         main="Heatmap of Correlation Matrix for j=5",
         cluster_rows=FALSE, cluster_cols=FALSE)
```



The correlation matrix for subject $j = 5$ shows predominantly positive correlations (red), especially within diagonal blocks, reflecting modular brain organization.

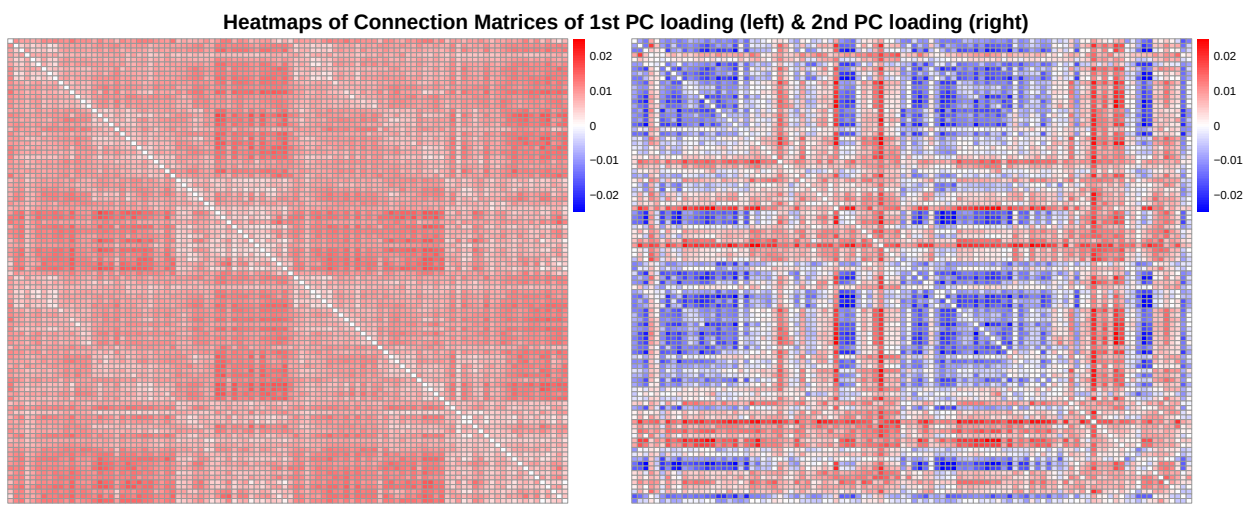
We then apply PCA to the $90 \times 10{,}000$ matrix:

```r
pca <- prcomp(vectorized_cor_matrix, center = TRUE, scale. = FALSE)
pcloading <- t(pca$rotation)
```

Reshaping PC loadings into 100×100 matrices reveals how ROI-to-ROI connections contribute to each principal component.

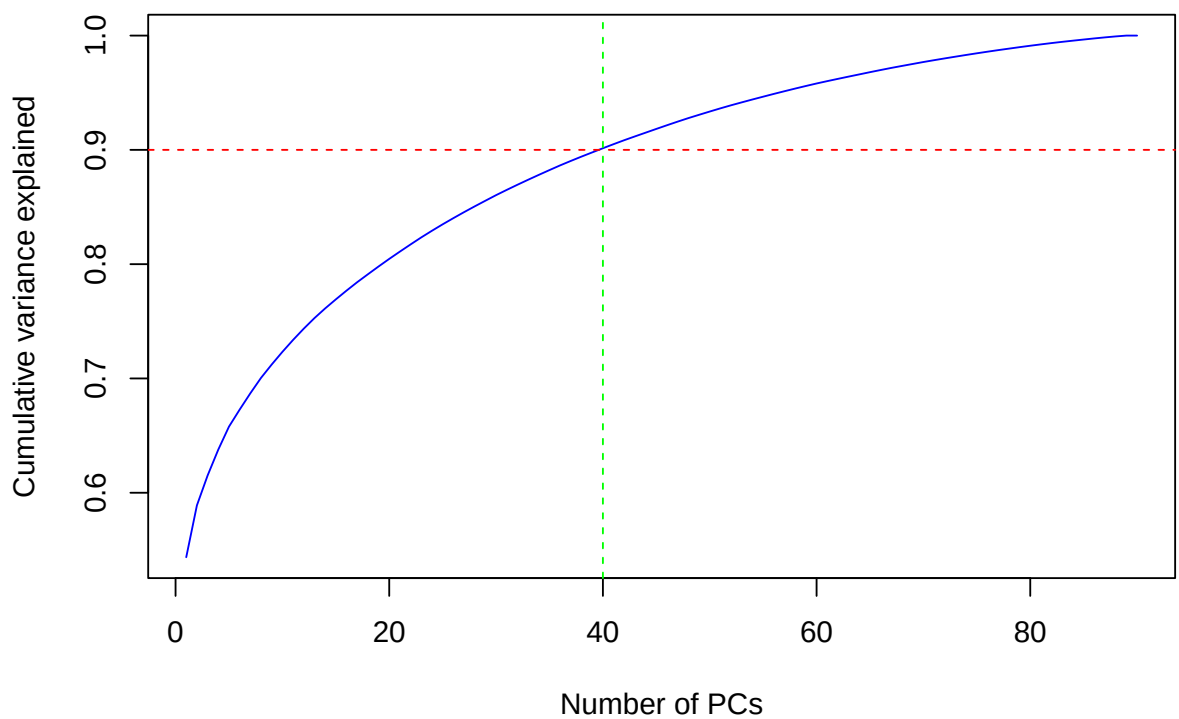
```r
V1 <- pcloading[1, ]
V2 <- pcloading[2, ]
PC1_cor <- matrix(V1, nrow = 100, ncol = 100) # to reconstruct matrix with same dimension as the
↪  correlation matrix
PC2_cor <- matrix(V2, nrow = 100, ncol = 100)

#dim(PC1_cor)
```

```r
# Create the heatmaps and capture the grobs
h1 <- pheatmap(PC1_cor,
               color = color_palette,
               breaks = seq(-0.025, 0.025, length.out = 101),
               cluster_rows = FALSE,
               cluster_cols = FALSE,
               silent = TRUE)

h2 <- pheatmap(PC2_cor,
               color = color_palette,
               breaks = seq(-0.025, 0.025, length.out = 101),
               cluster_rows = FALSE,
               cluster_cols = FALSE,
               silent = TRUE)

# Directly print the combined plot
grid.arrange(grobs = list(h1[[4]], h2[[4]]),
             nrow = 1,
             top = textGrob("Heatmaps of Connection Matrices of 1st PC loading (left) & 2nd PC
             ↪  loading (right)",
                            gp = gpar(fontsize = 18, fontface = "bold")))
```



PC1 mainly displays positive weights (red), indicating widespread co-variation across connections, while PC2 shows both positive and negative contributions, reflecting contrasting network patterns. Strongest loadings cluster within blocks, corresponding to functionally connected regions.

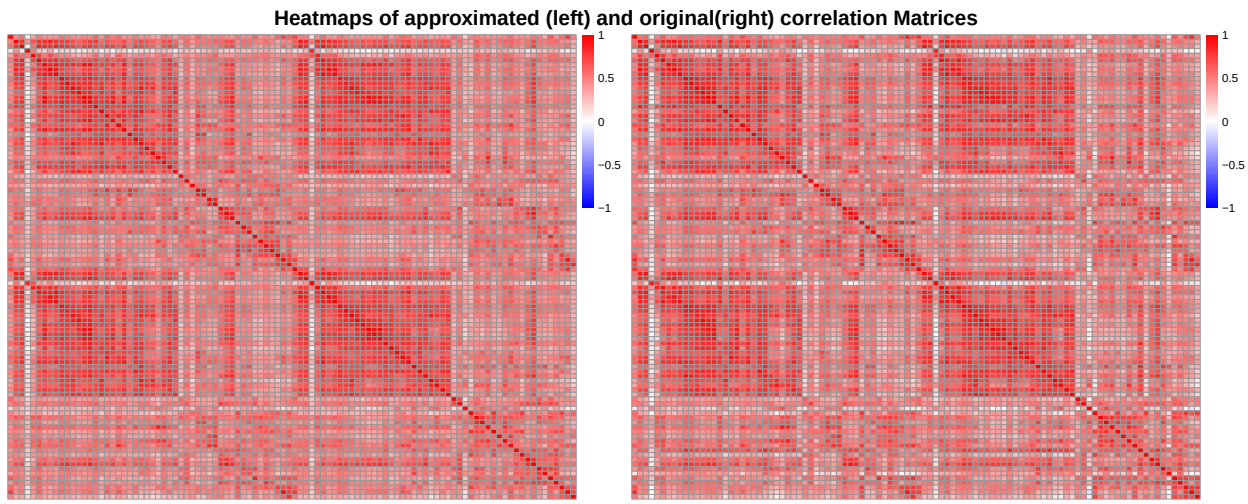
```r
# Variance explained
cumsumvar <- cumsum(pca$sdev^2) / sum(pca$sdev^2)
plot(cumsumvar, type="l", col="blue", xlab="Number of PCs",
     ylab="Cumulative variance explained")
abline(h=0.9,col="red",lty=2)
abline(v=which(cumsumvar>=0.9)[1],col="green",lty=2)
```



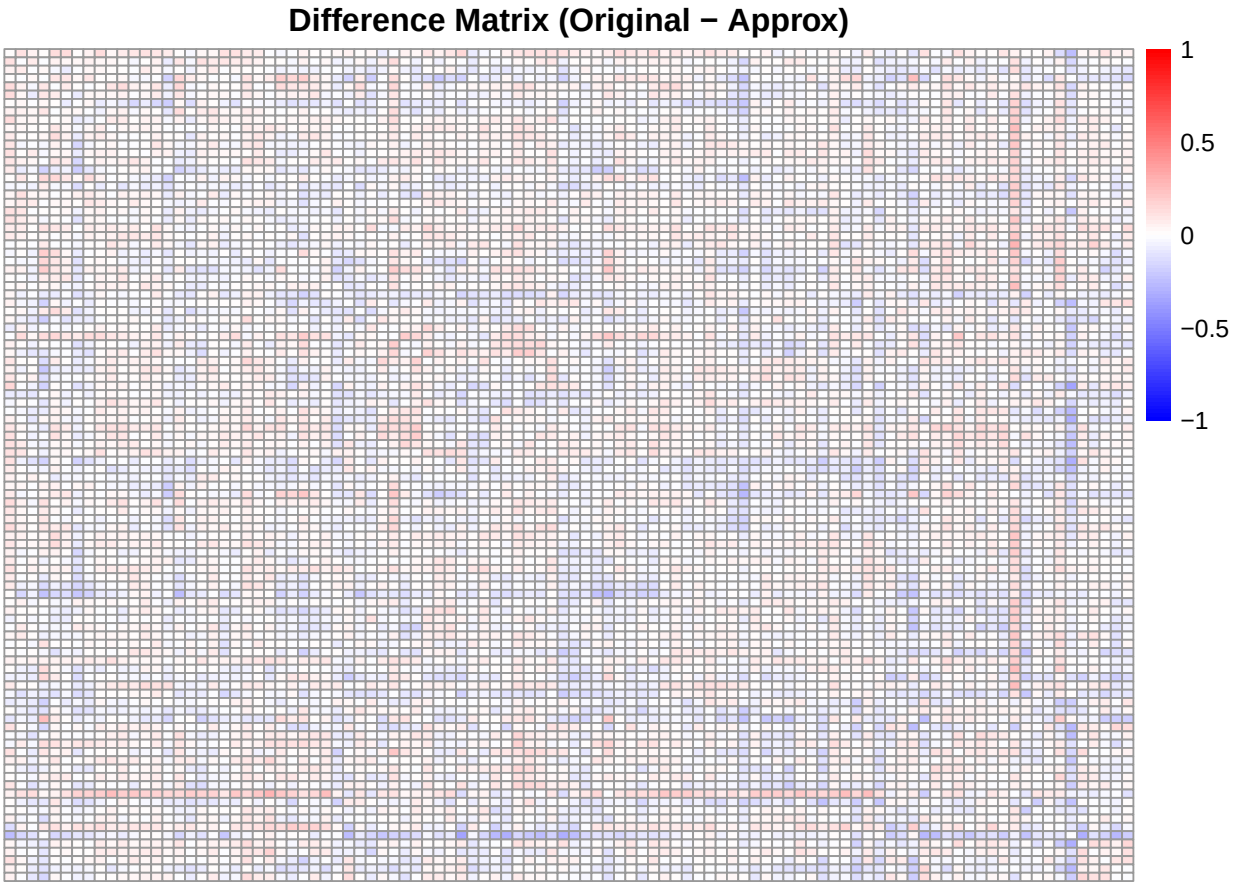
The first 40 components explain ~ 90 of the total variance. Using these, we reconstruct the correlation matrix of subject $j = 5$:

```r
# function to reconstruct data for correlation matrix j with rank r (number of components)
approx_data <- function(pca, j, r){
  reconst <- pca$center + pca$x[j, 1:r] %*% t(pca$rotation[, 1:r])
  return(reconst)
}


# Compute the approximated matrix
x.approx <- matrix(approx_data(pca, 5, 40), nrow = 100, ncol = 100)

# Compute the difference between original and approximated matrices
diff_matrix <- corr_5 - x.approx

# Determine the scale limits (centered at zero)
breaks <- seq(-1, 1, length.out = 101)  # Ensure white is at zero

# Plot the approximated correlation matrix
plot_approx_correlation = pheatmap(x.approx,
```

```r
         color = color_palette,
         breaks = breaks,  # Set scale to be symmetric around 0
         cluster_rows = FALSE,
         cluster_cols = FALSE,
         silent = TRUE)

# Plot of the original correlation matrix
plot_original_correlation = pheatmap(corr_5,
         color = color_palette,
         breaks = breaks,
         cluster_rows = FALSE,
         cluster_cols = FALSE,
         silent = TRUE)

# Combine the plots side by side
grid.arrange(grobs = list(plot_approx_correlation[[4]], plot_original_correlation[[4]]),
             nrow = 1,
             top = textGrob("Heatmaps of approximated (left) and original(right) correlation
             ↪  Matrices",
                            gp = gpar(fontsize = 18, fontface = "bold")))
```




```r
# Plot the difference matrix
pheatmap(diff_matrix,
         color = color_palette,
         main = "Difference Matrix (Original - Approx)",
         breaks = breaks,
         cluster_rows = FALSE,
         cluster_cols = FALSE)
```



The reconstructed and original matrices are visually similar, confirming that PCA effectively preserves the main connectivity structure. This yields a compact 90×40 representation summarizing inter-subject variability in brain connectivity.

Overall, correlation-based PCA allows consistent dimensionality reduction across subjects while retaining essential network-level patterns.

4 Advanced Statistical Analysis – Co-Activation Patterns (CAP)

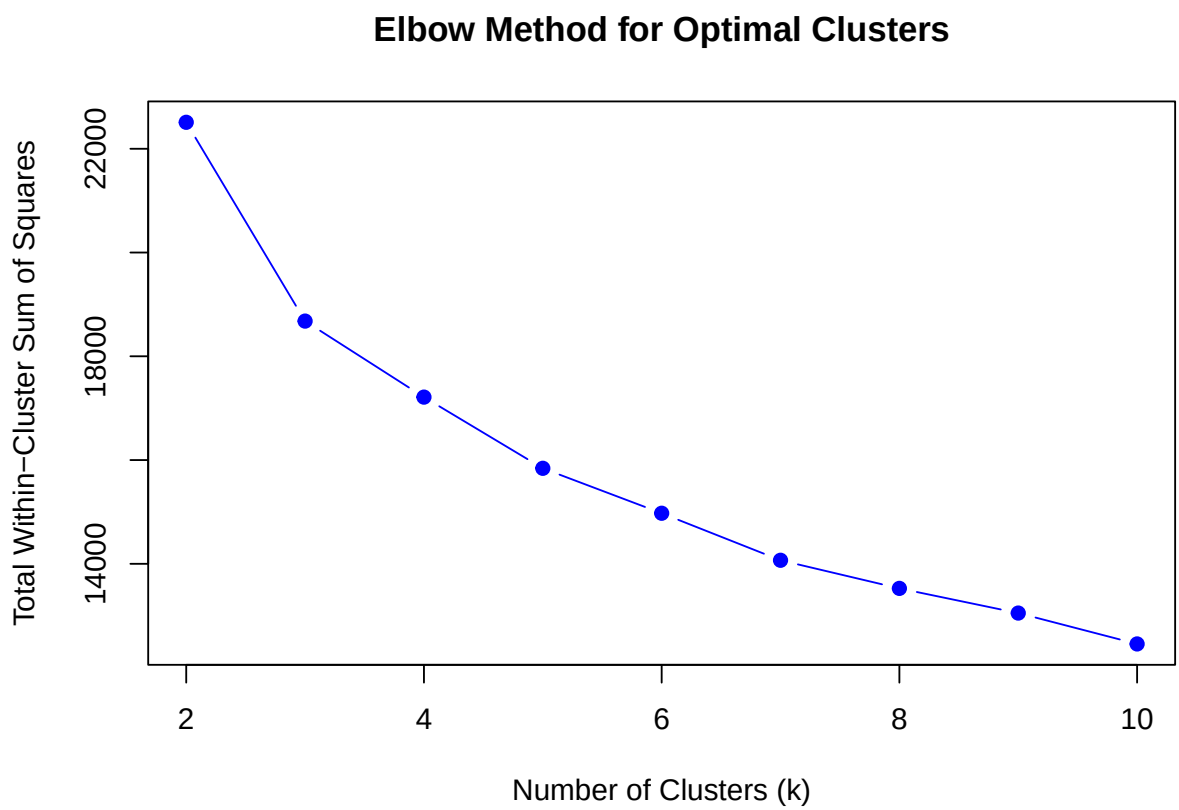
We apply Co-Activation Pattern (CAP) analysis to subject $j = 5$, following Liu et al. (2018). Unlike Static Functional Connectivity (FC), which provides a time-averaged view of brain correlations, CAP captures **dynamic** states of co-activation by clustering time points showing similar activation patterns. Each CAP thus represents a distinct brain state.

To identify high-activity moments, we select the 30% most active time points based on the absolute mean activation across ROIs (threshold = 70th percentile). Taking absolute values retains both activations and deactivations, which reflect neural resource allocation (Kudo et al., 2004).

```
# Select top 30% most active time points for subject 5
mean_activation <- colMeans(abs(subject_5_data))
threshold <- quantile(mean_activation, 0.70)
selected_time_points <- which(mean_activation > threshold)
filtered_data <- subject_5_data[, selected_time_points]
```

We cluster these time points using k-means and determine the optimal number of clusters k using the elbow method.

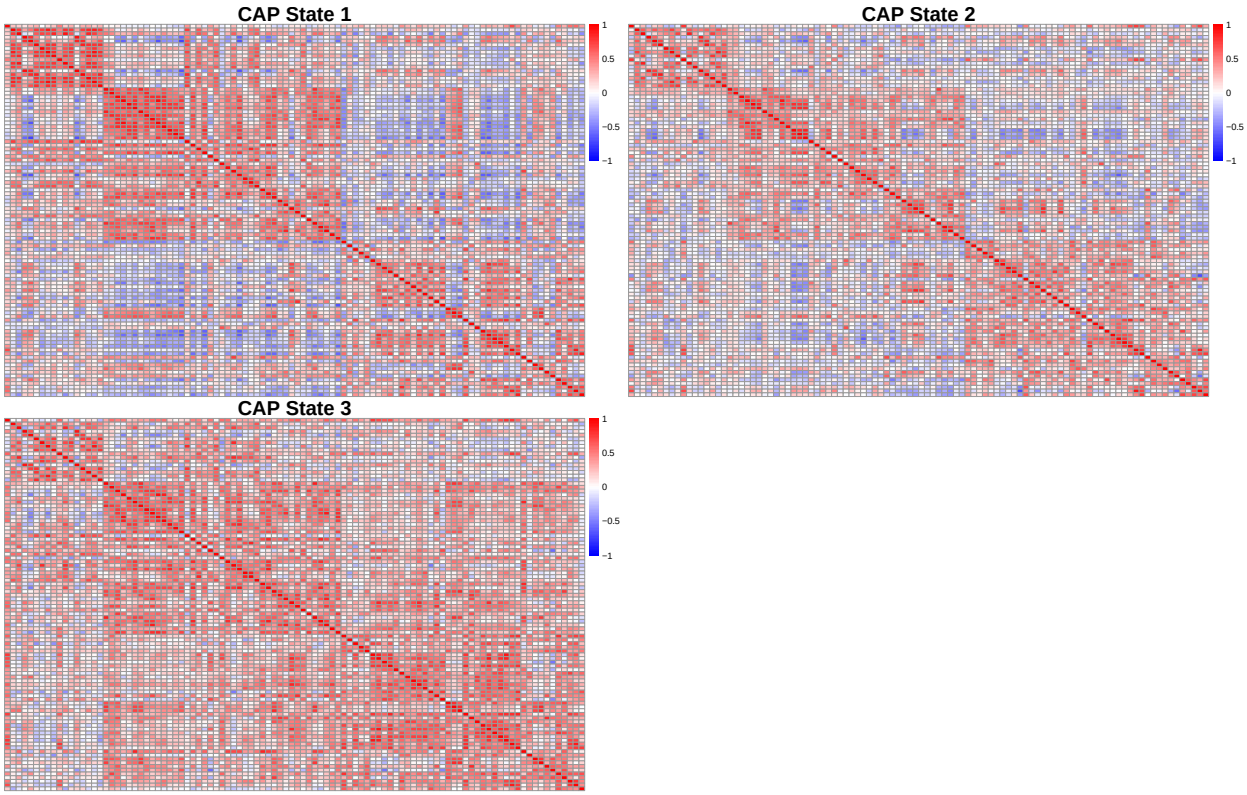
```r
# Define range of k values to test
k_values <- 2:10

# Initialize vector to store WCSS values
wcss <- numeric(length(k_values))

# Loop over k values, perform k-means, and store total within-cluster sum of squares
for (i in seq_along(k_values)) {
  set.seed(123)  # For reproducibility
  result <- kmeans(t(filtered_data), centers = k_values[i], nstart = 10)
  wcss[i] <- result$tot.withinss
}

# Simple elbow plot
plot(k_values, wcss, type = "b", pch = 19,col = "blue",
     xlab = "Number of Clusters (k)",
     ylab = "Total Within-Cluster Sum of Squares",
     main = "Elbow Method for Optimal Clusters")
```



There is not a clear drop in elbow plot. We decide to choose k=3 as the following drops are less steep than the one from 2 to 3.

```r
# Set seed and number of clusters
set.seed(11)

# Create new ROI order by sorting indices based on network_labels

# Determine the scale limits (centered at zero)
```

```r
breaks <- seq(-1, 1, length.out = 101)  # Ensure white is at zero

# Names of the different networks
network_names <- c("Visual", "Somatomotor", "Dorsal Attention", "Salience", "Limbic", "Central
↪  Executive", "Default")

# Functional networks (ROI indices labelled with corresponding network)
network_labels <- c(rep(network_names[1], 9), rep(network_names[2], 6), rep(network_names[3], 8),
↪  rep(network_names[4], 7),
                    rep(network_names[5], 3), rep(network_names[6], 4), rep(network_names[7], 13),
                    rep(network_names[1], 8), rep(network_names[2], 8), rep(network_names[3], 7),
                    ↪  rep(network_names[4], 5),
                    rep(network_names[5], 2), rep(network_names[6], 9), rep(network_names[7], 11))

new_order <- order(factor(network_labels, levels = network_names))

k <- 3
cap_clusters <- kmeans(t(filtered_data), centers = k, nstart = 10)

# Store heatmaps
cap_heatmaps <- list()

for (i in 1:k) {
  # Time points belonging to this CAP state
  cap_data <- filtered_data[, cap_clusters$cluster == i]

  # Correlation matrix
  cor_matrix <- cor(t(cap_data))
  cor_matrix <- cor_matrix[new_order, new_order]  # Reorder

  # Create heatmap as grob (silent = TRUE to store instead of display)
  hm <- pheatmap(cor_matrix,
                 main = paste("CAP State", i),
                 cluster_rows = FALSE,
                 cluster_cols = FALSE,
                 color = color_palette,
                 breaks = breaks,
                 silent = TRUE)

  #Agrandir le titre
  hm$gtable$grobs[[1]]$gp <- gpar(fontsize = 22, fontface = "bold")

  cap_heatmaps[[i]] <- hm$gtable
}

# Display heatmaps together in a grid (e.g., 1 row of 3 columns)
grid.arrange(grobs = cap_heatmaps, nrow = 2)
```

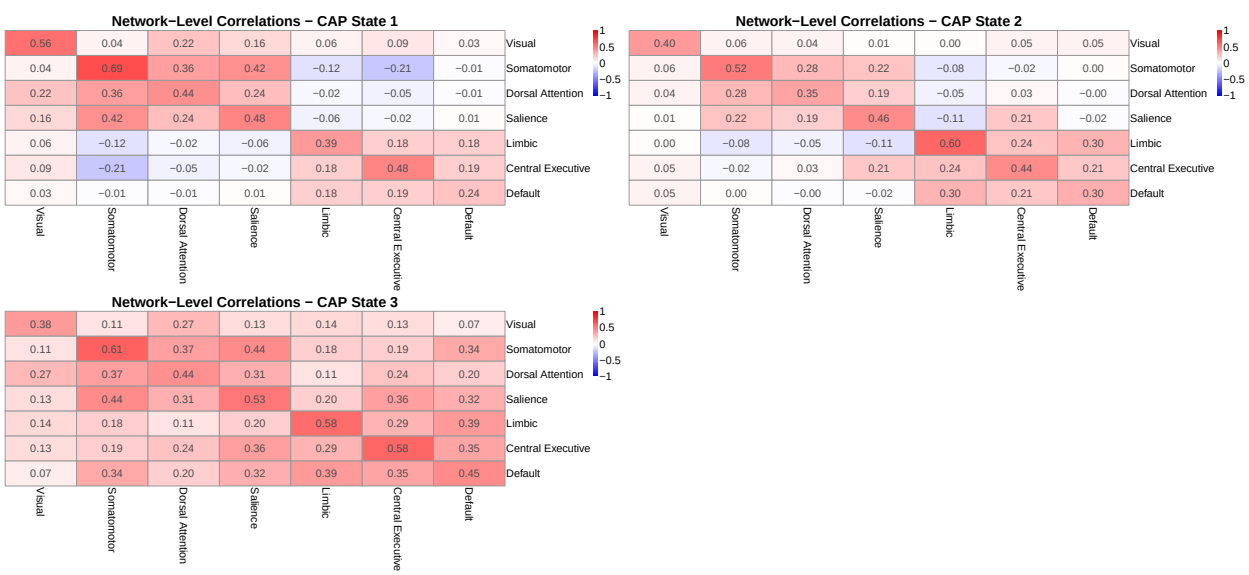


Each CAP is represented by a correlation matrix summarizing how ROIs co-activate within that state. CAP 1 and CAP 2 display similar structures: strong positive correlations within networks (Somatomotor, Dorsal Attention, Salience) and weaker or negative correlations with others. CAP 1 shows stronger within-network connectivity, suggesting task engagement or alertness, while CAP 2 presents higher correlations among Central Executive, Limbic, and Default networks—possibly linked to internal or emotional processing. CAP 3 exhibits globally positive correlations across networks, consistent with transient large-scale co-activations described by He and Liu (2012).

To better visualize interactions, we average correlations within and between the seven functional networks identified earlier.

```
# Create a list to store all the heatmaps
heatmap_list <- list()

# Loop through each CAP state
for (i in 1:k) {
  # Select time points that belong to the i-th CAP state
  cap_data <- filtered_data[, cap_clusters$cluster == i]

  # Compute the ROI-level correlation matrix
  cor_matrix <- cor(t(cap_data))

  # Initialize the network-level correlation matrix
  unique_networks <- unique(network_labels)
  network_corr <- matrix(0, nrow = length(unique_networks), ncol = length(unique_networks))
  rownames(network_corr) <- colnames(network_corr) <- unique_networks

  # Fill in the network-level correlation matrix by averaging ROI correlations
  for (net1 in unique_networks) {
    for (net2 in unique_networks) {
      idx1 <- which(network_labels == net1)
```

```r
      idx2 <- which(network_labels == net2)
      network_corr[net1, net2] <- mean(cor_matrix[idx1, idx2])
    }
  }

  # Generate the heatmap (stored as a grob object)
  p <- pheatmap(
    network_corr,
    main = paste("Network-Level Correlations - CAP State", i),
    cluster_rows = FALSE,
    cluster_cols = FALSE,
    color = colorRampPalette(c("blue", "white", "red"))(100),
    breaks = breaks,
    display_numbers = TRUE,
    silent = TRUE,
    fontsize = 25,              # General font size
    fontsize_number = 25,       # Font size for numbers inside cells
    fontsize_row = 25,          # Font size for row labels
    fontsize_col = 25           # Font size for column labels
  )

  # Store the plot as a grob in the list
  heatmap_list[[i]] <- p$gtable
}

# Display all heatmaps in a 2-column grid
grid.arrange(grobs = heatmap_list, ncol = 2)
```



Across all CAPs, within-network correlations remain high, confirming correct ROI labeling and functional coherence. Differences across CAPs highlight dynamic shifts between externally oriented (task-engaged) and internally oriented (rest-like) brain states.

Limitations include the arbitrary activity threshold (70th percentile), the assumption of independent time points (no temporal continuity checked), and sensitivity to the number of clusters k. Despite these, CAP analysis provides an interpretable low-dimensional view of dynamic brain states and complements the static connectivity analysis performed earlier.

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